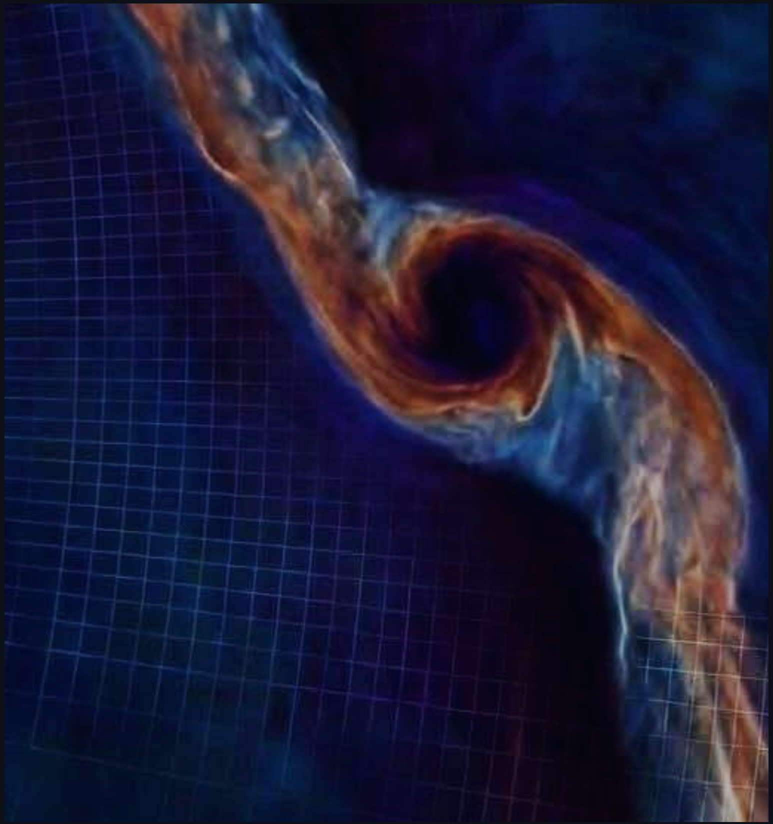
# Not just gas:

## How solids-driven torques shaped the migration of the Galilean moons

**Leonardo Krapp**

3rd PFITS+ Meeting on Hydrodynamics of Circumstellar Disks and Planet Formation



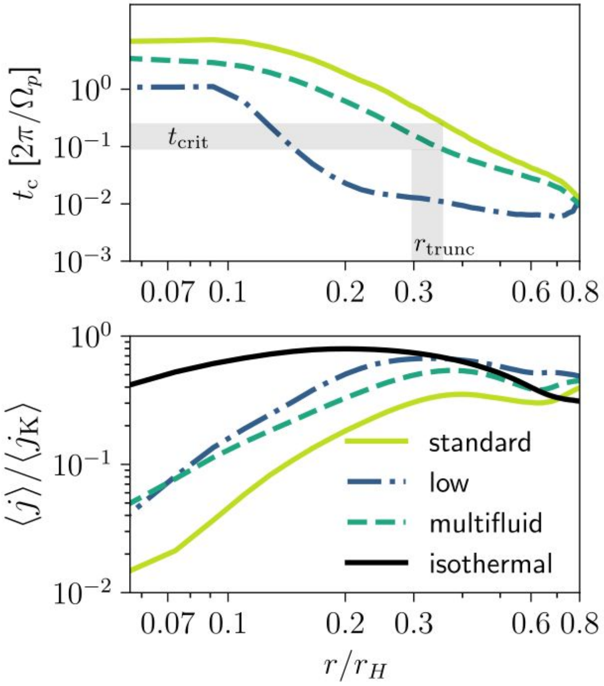
FARGO3D: MULTIFLUID CIRCUMPLANETARY DISK IN NON UNIFORM MESHES WITH RAPID ADVECTION (RAM)

HEISING-SIMONS FOUNDATION

Universidad de Concepción Chile

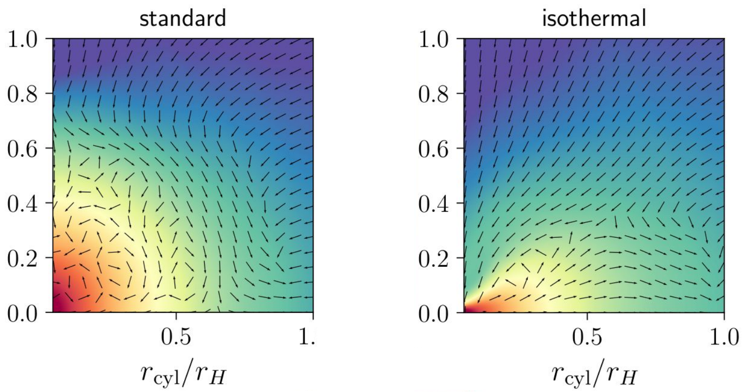
# DUST-DRIVEN COOLING IS CRITICAL FOR CPD FORMATION

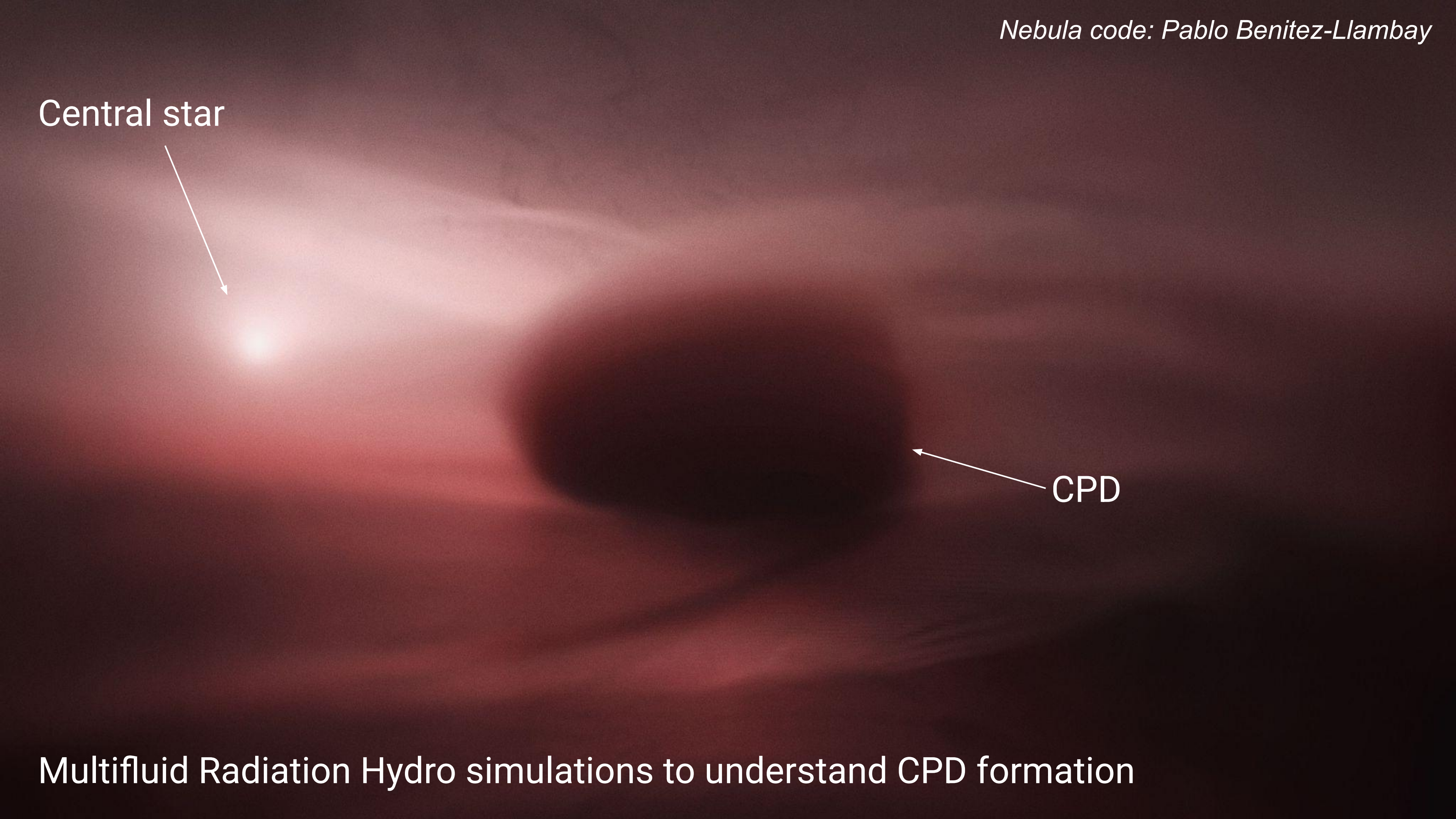
Krapp et al. 2024




Approximate cooling condition
based on 3D radiation hydrodynamics

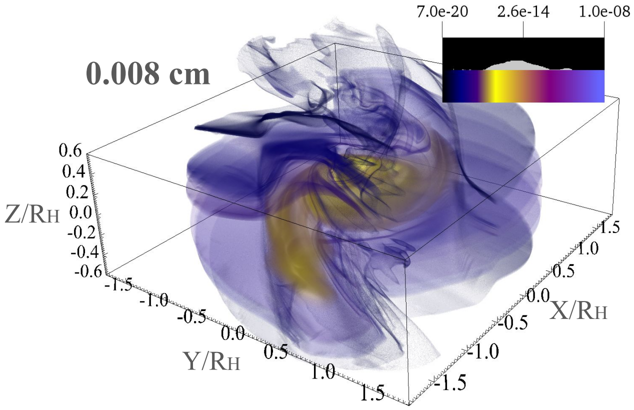
$$\frac{t_c}{2\pi/\Omega_p} \gtrsim 0.1 \implies \langle j\rangle \lesssim \langle j_{\rm iso}\rangle$$

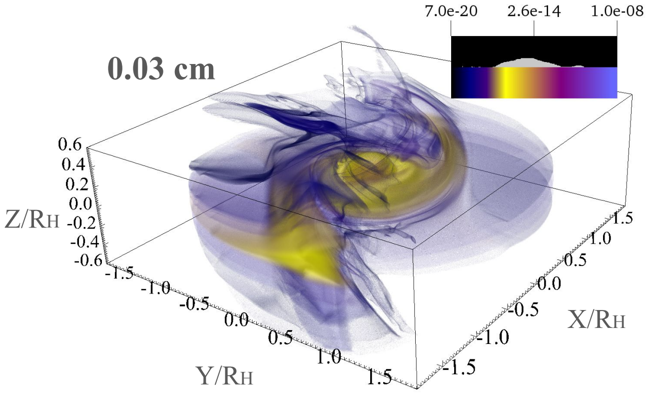


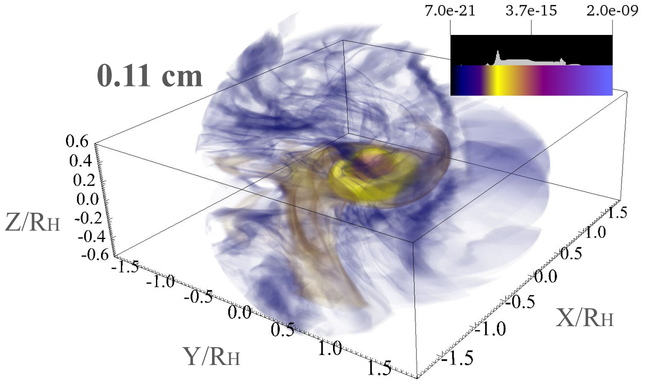

Nebula code: Pablo Benitez-Llambay
Central star
CPD
Multifluid Radiation Hydro simulations to understand CPD formation

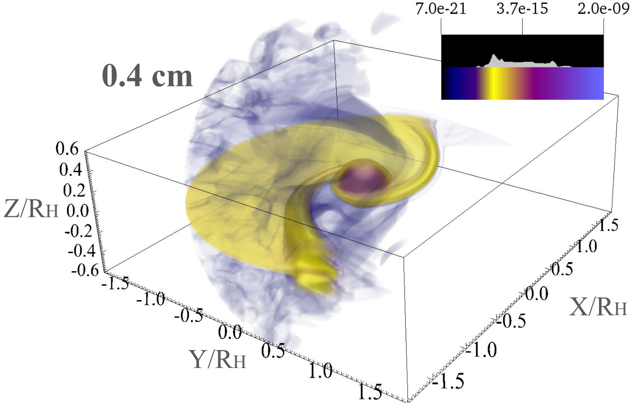
# Let’s take a look at the solid material.

**Aiden Kriel**
PhD candidate
Northern Arizona University



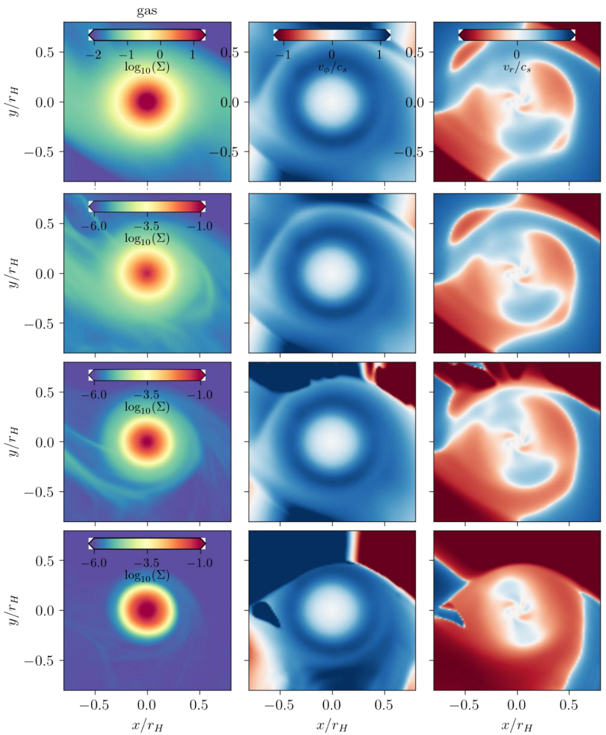



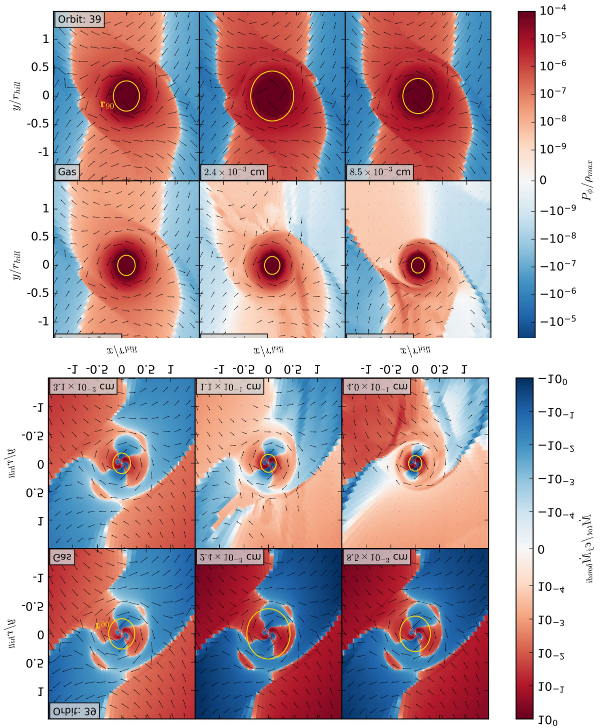



# Particles are not well-mixed with gas
# Solid subdisk are Compact & Optically thick

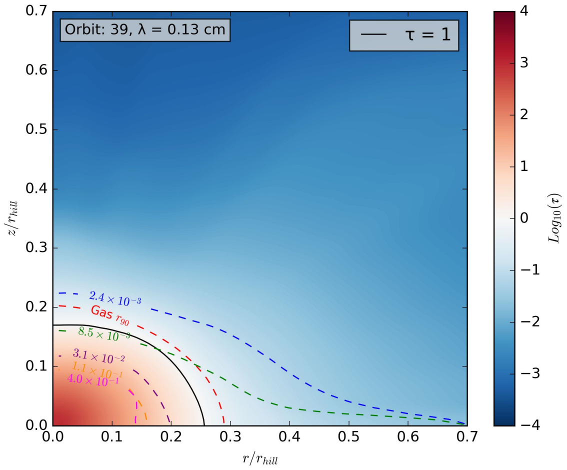
## Compact subdisk $\ell$ < Rhill/3




## High Mass flux at the midplane

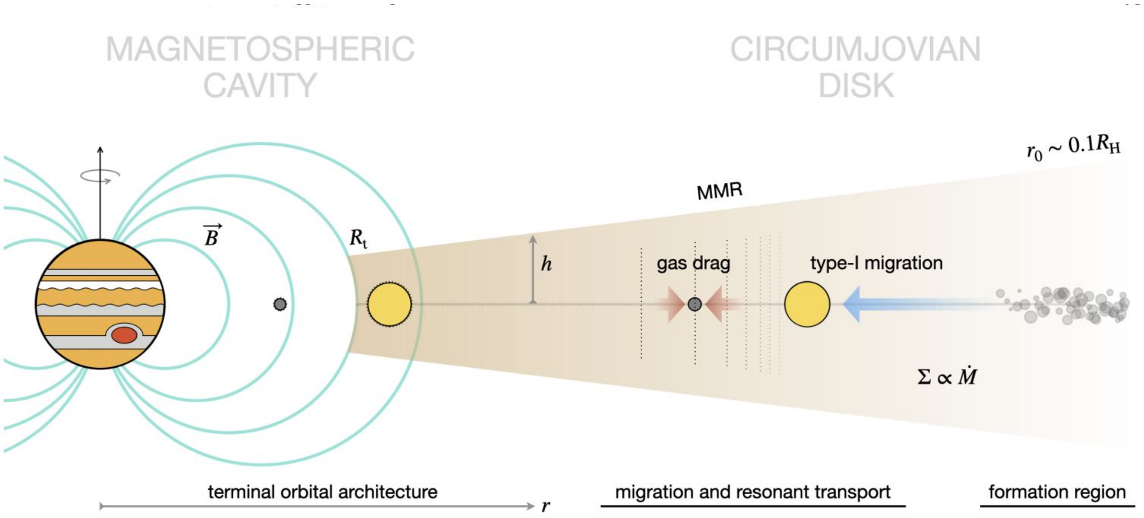



## Optically thick subdisk




Kriel et al. (in prep)

# Galilean Moons: Formation in the Circumjovian disk



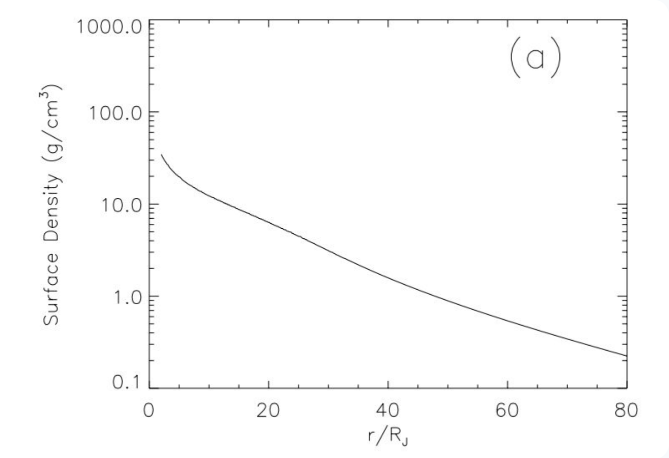

Brunton & Batygin (2025)

# Classical Models of Circumjovian Disks

- **Orbital Migration can be fast for moon survival**

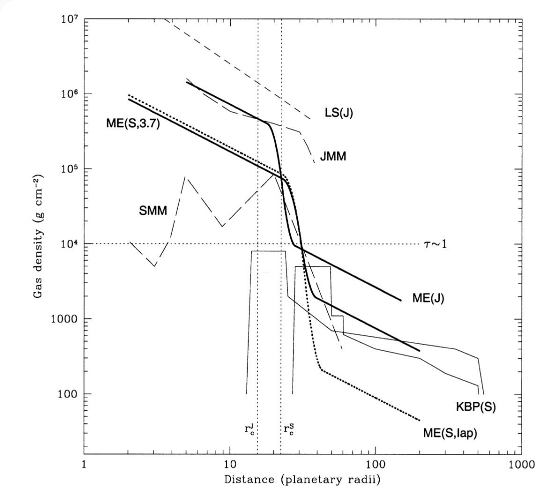
## Gas-starved disk model

Canup & Ward (2002)




## Minimum mass subnebula

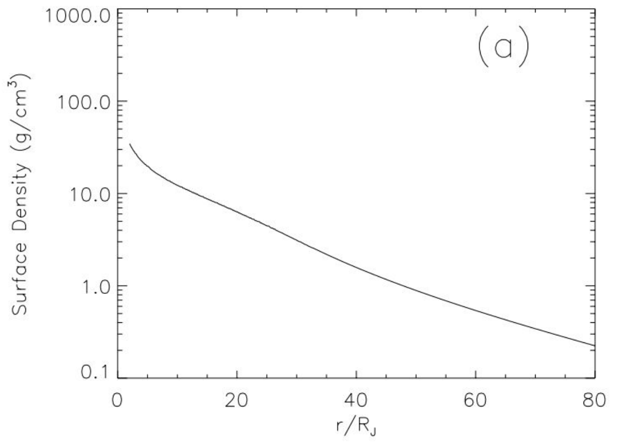
Mosqueira & Estrada (2002)



# Classical Models of Circumjovian Disks

- Orbital Migration can be fast challenging moon survival
- Enhanced solid-to-gas ratios (Z >0.03 )

### Gas-starved disk model
Canup & Ward (2002)




### Minimum mass subnebula
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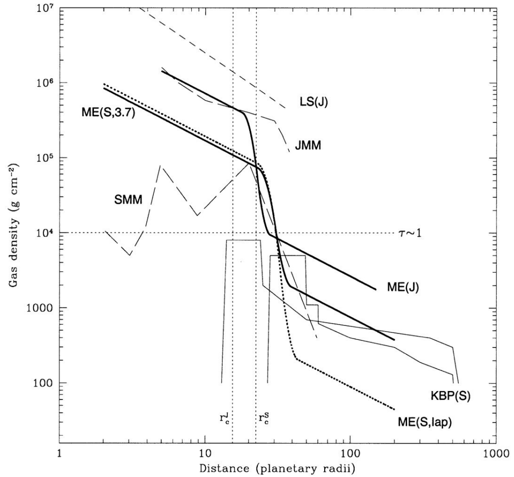
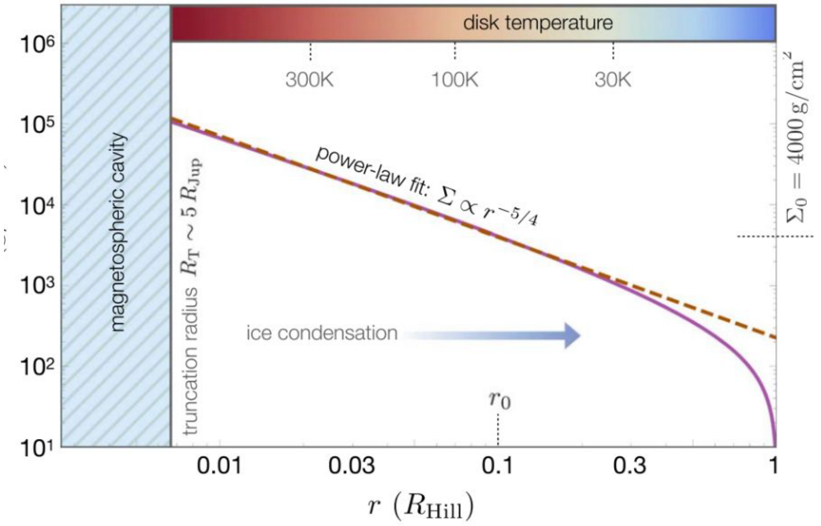
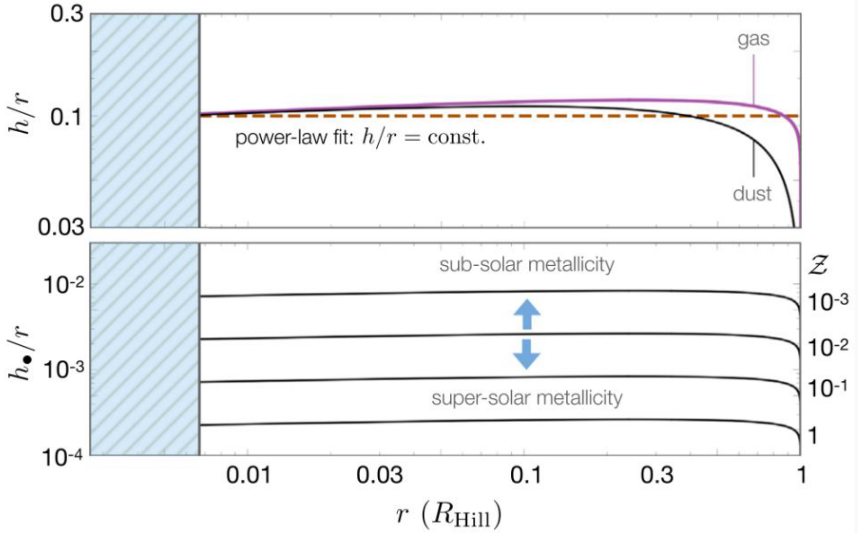


# Decretion Disk Model

## Pebble and Planetesimal Accretion



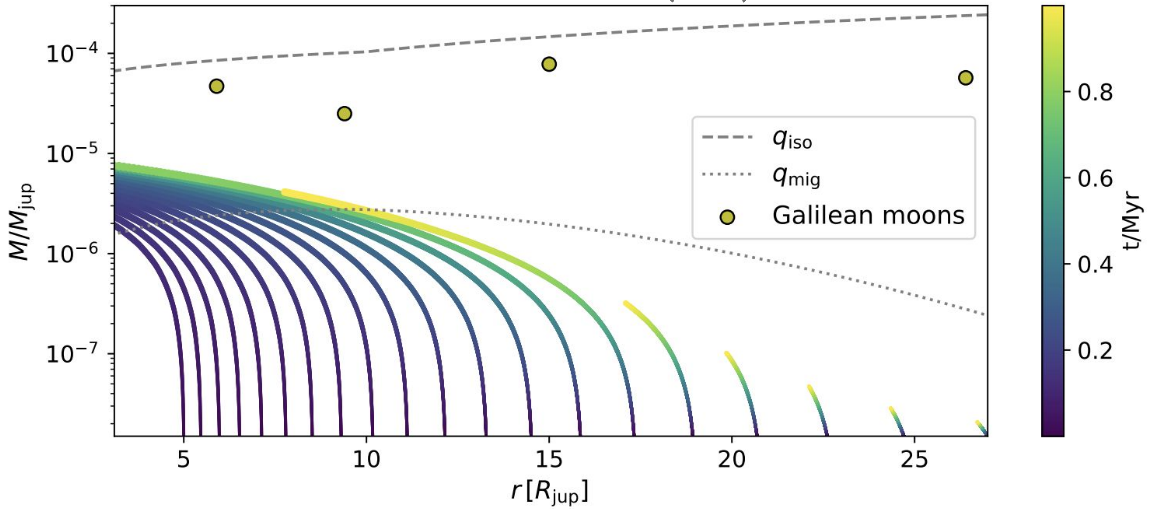

Batygin & Morbidelli (2020)

# Architects of the Galilean System

Evaluating the role of gas-solids dynamics in moon migration

## Migration is fast and a challenge for formation models

Ronnet & Johansen (2020)




Sasaki et al. 2010: **Migration stops at a inner truncation radius**

Ogihara & Ida 2012 (and more)

# Early Arrivals & Orbital Evolution

Can moons form while the disk is still young?

Do solids dictate their final orbits?

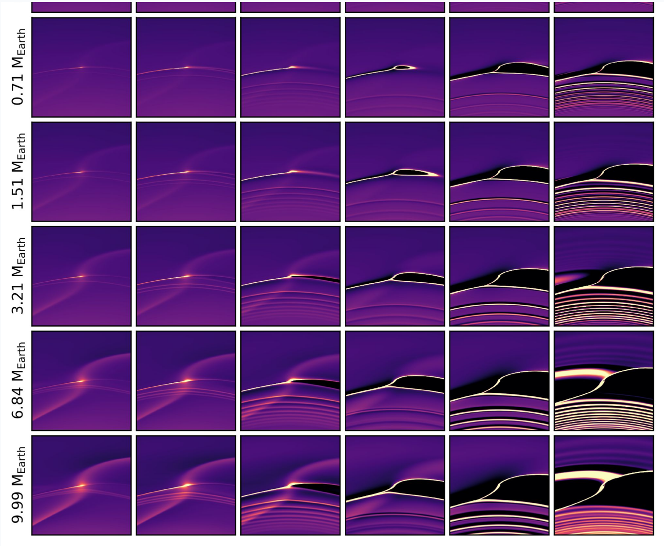
# Early Arrivals & Orbital Evolution

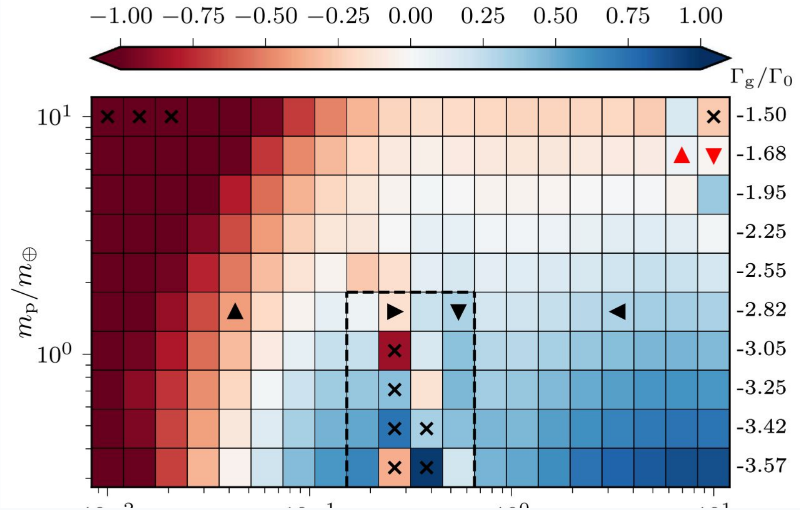
Can moons form while the disk is still young?

Do solids dictate the Galilean moons final orbits?

# Gas-Solids Dynamics and Planet Migration

## SOLIDS dynamics could favor OUTWARD migration



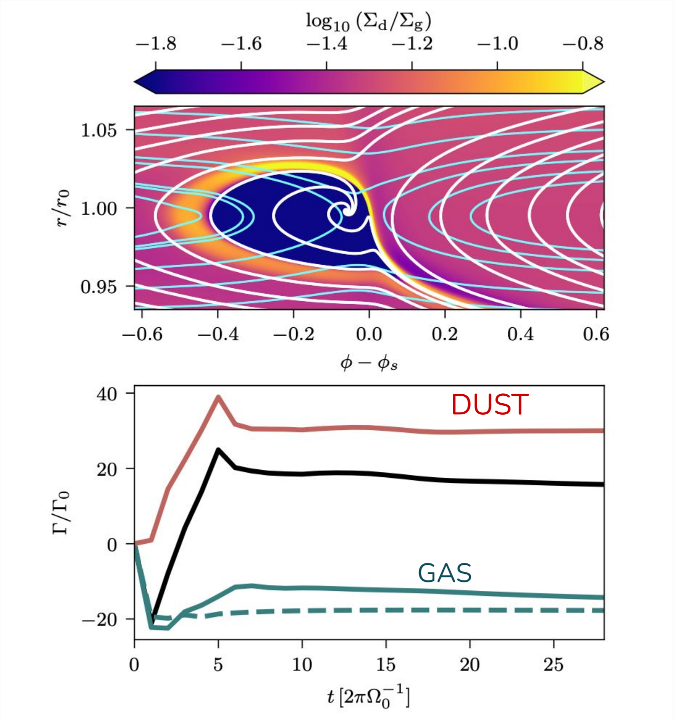




Source: Benitez-Llambay & Pessah (ApJL, 2018)

*See also Regály (2020)*

# Gas-Solids Migration in Moons

Lucas Gonzalez-Rivas et al. (2026, A&A Letter to Editor)




Lucas Gonzalez Rivas

Undergrad student at UdeC

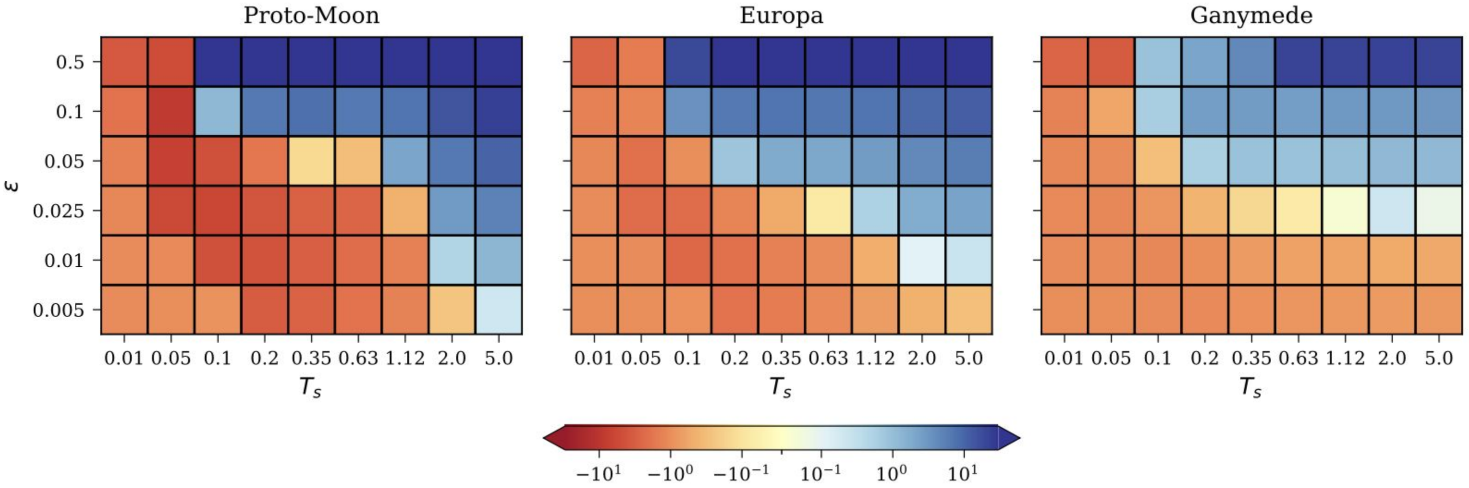
(2025, A&A L)

Looking for PhD positions

# 2D Two-fluid FARGO3D Simulations

Lucas Gonzalez-Rivas et al. (2026, A&A Letter to Editor)

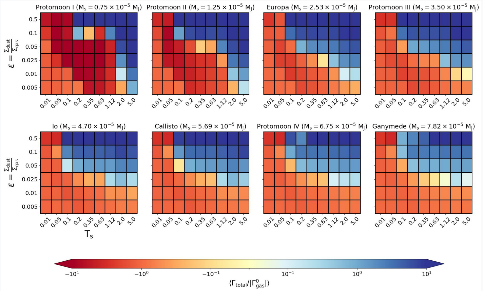
## Migration Map: Zero Eccentricity Orbit




**BLUE: OUTWARD MIGRATION TREND**

# REBOUND + DUST TORQUE MODEL

## Lucas Gonzalez-Rivas et al. in prep.




*Cube of models: Mass, dust-to-gas ratio, and Stokes number*

### Evolutionary tracks